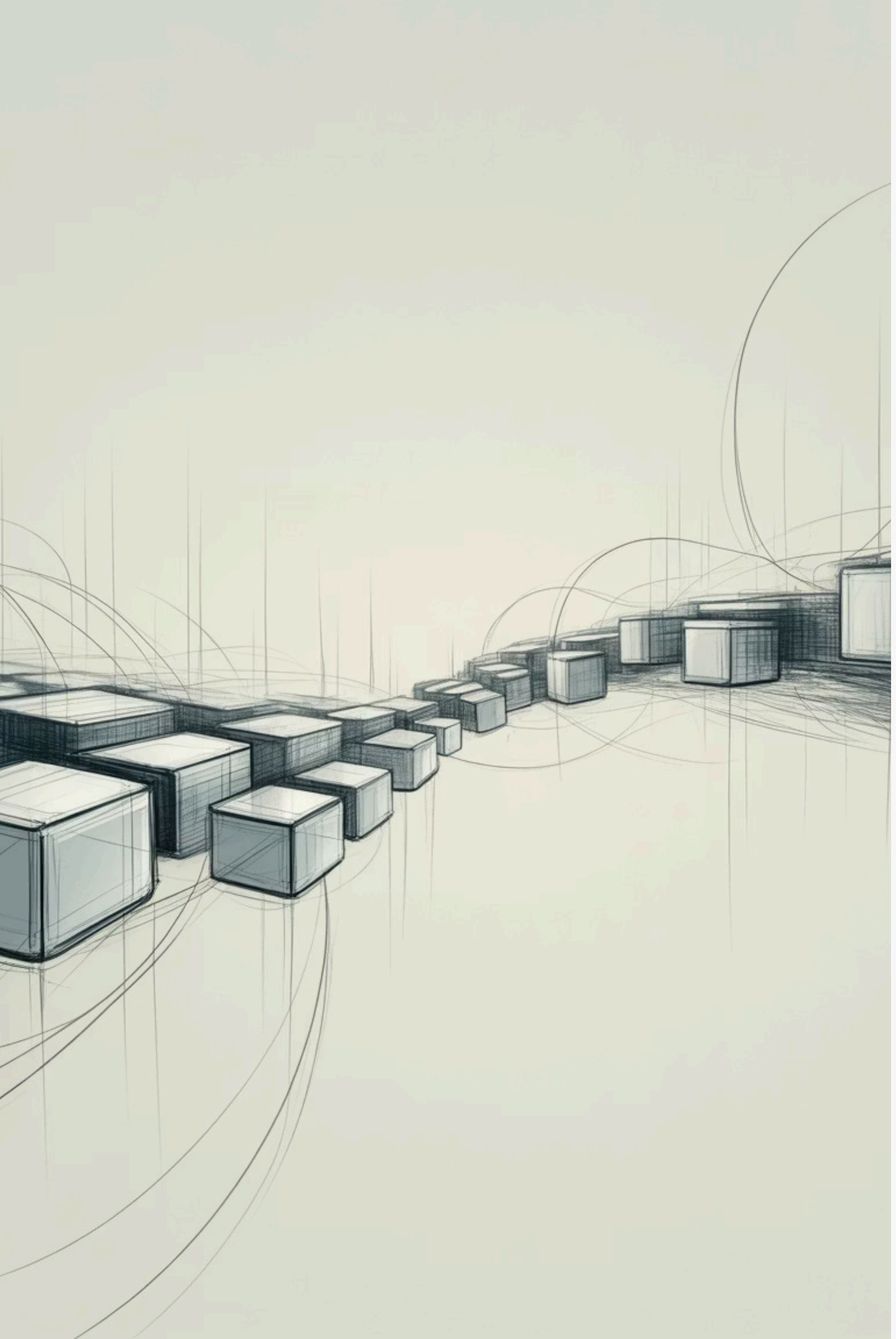
Kubernetes: The Engine Behind Modern Applications

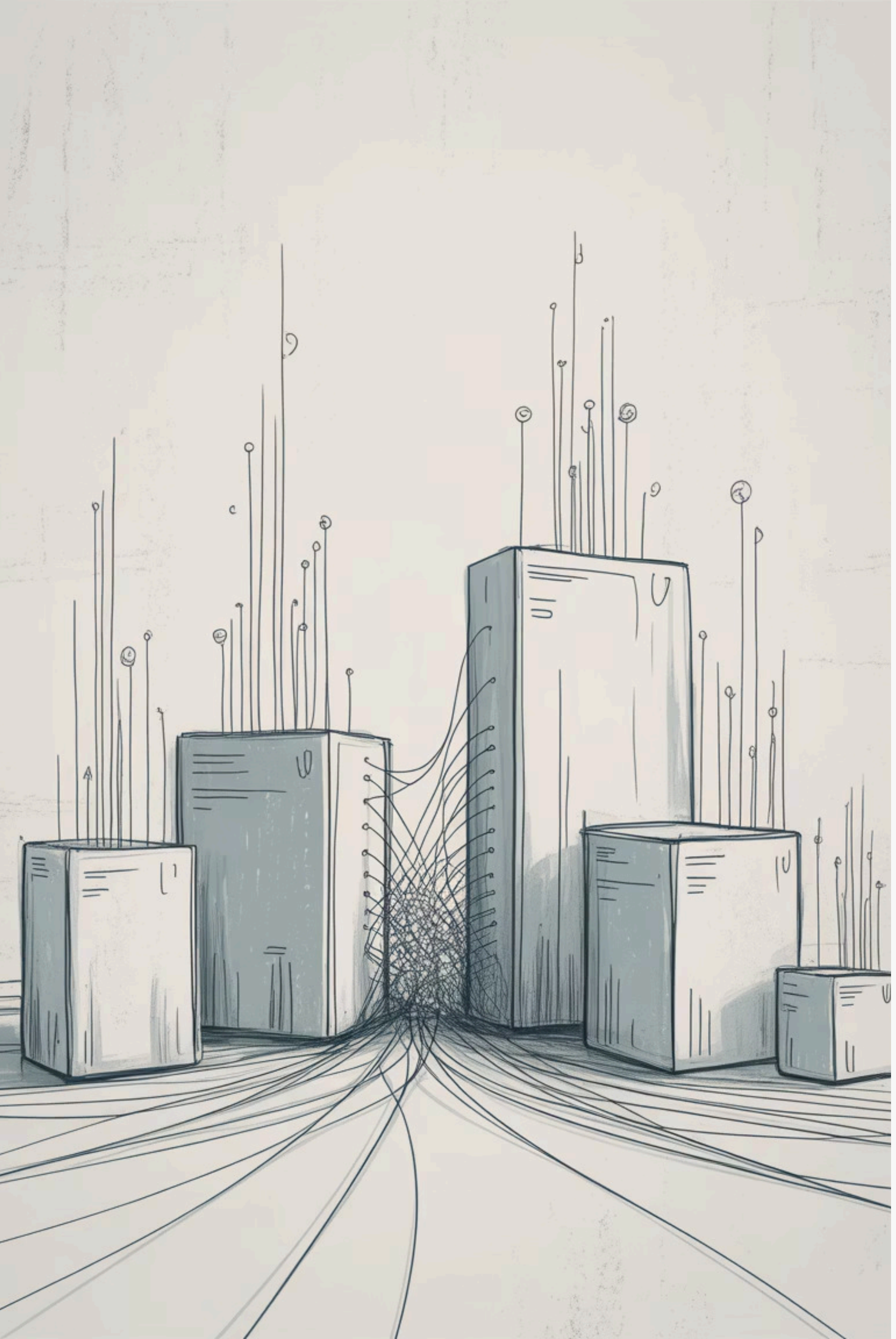
A Beginner-Friendly Introduction

by Naresh Chaurasia



Why Kubernetes?

- Applications run everywhere: cloud, data centers, hybrid
- Need faster deployments & scalability
- Kubernetes automates running apps in containers



What is Kubernetes?

- Open-source platform
- Manages containers (like Docker)
- Handles deployment, scaling, maintenance
- Think of it as an "OS for the cloud"

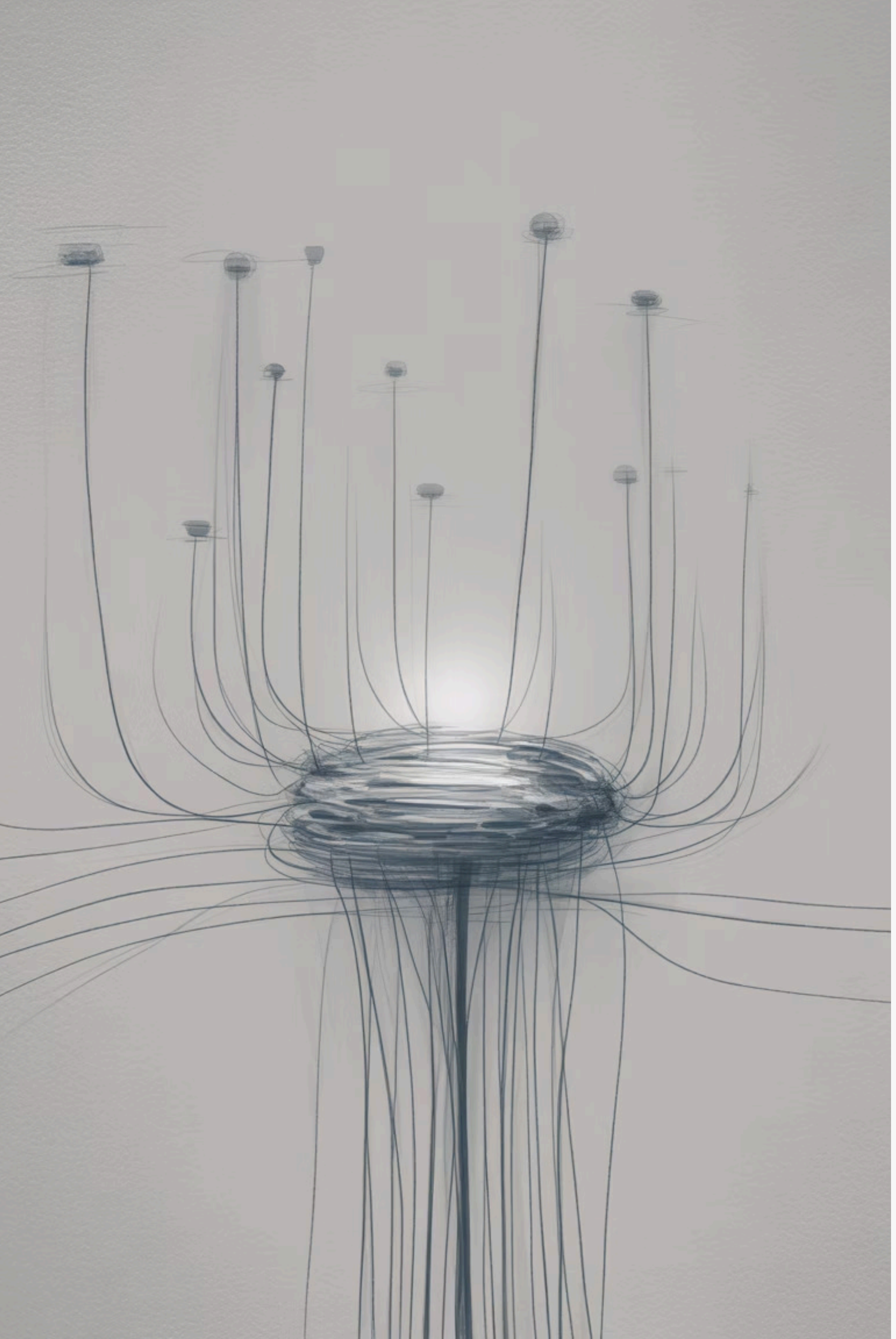
Key Features

Automatic Scaling

Self-Healing

Load Balancing

Portability



Core Components (Simple View)

Pod: Smallest unit (app + dependencies)

Node: Machine running pods

Cluster: Group of nodes

Control Plane: Brain that manages everything

Real-World Analogy

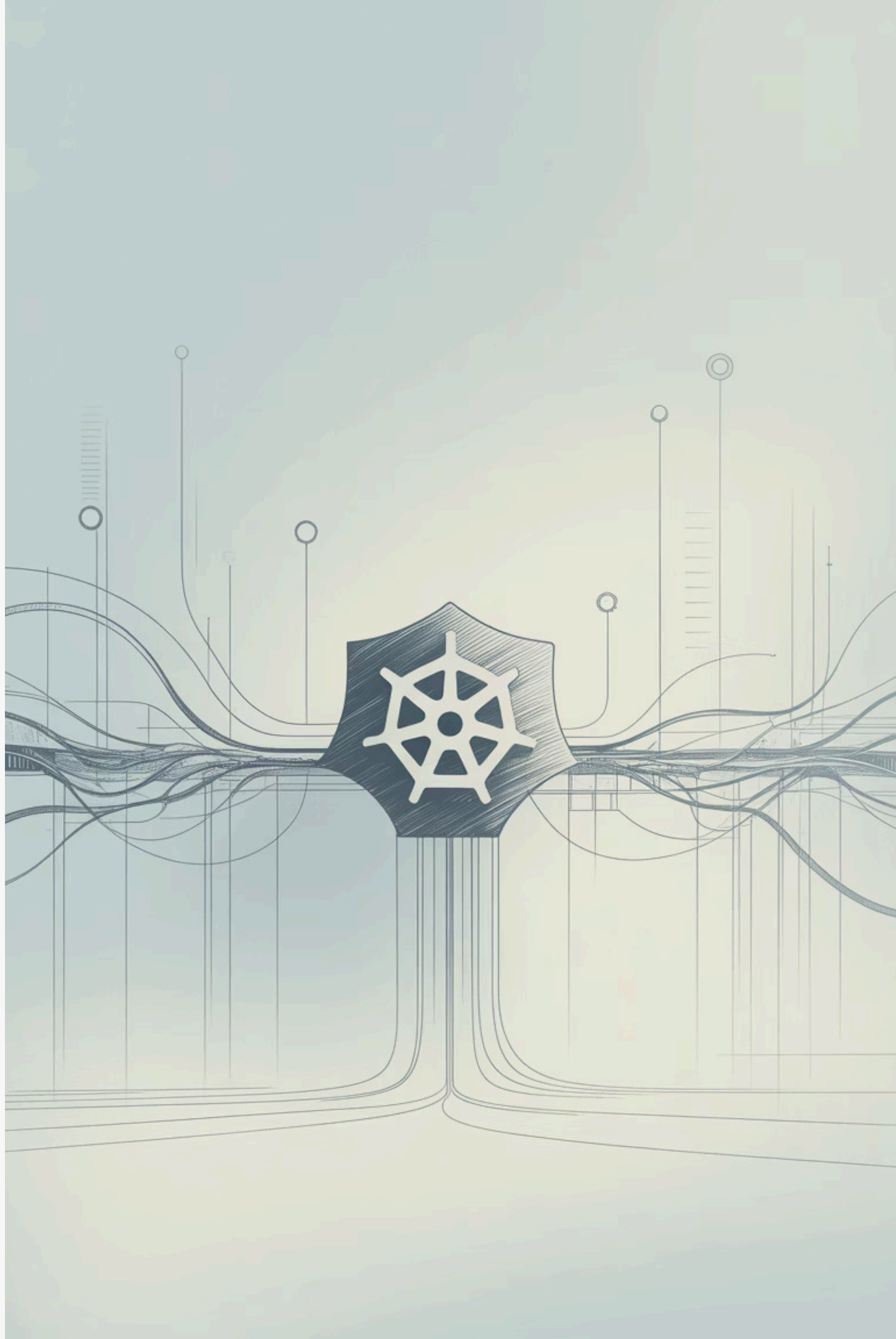
- *Pod* = Cargo box
- *Node* = Ship
- *Cluster* = Fleet
- *Control Plane* = Port authority

Kubernetes in Companies

Used by Google, Netflix, Spotify, Airbnb

Enables multiple updates per day

Saves cost & improves reliability



Kubernetes Workflow (Simplified)

01

Developer makes container

02

Deployed to Kubernetes

03

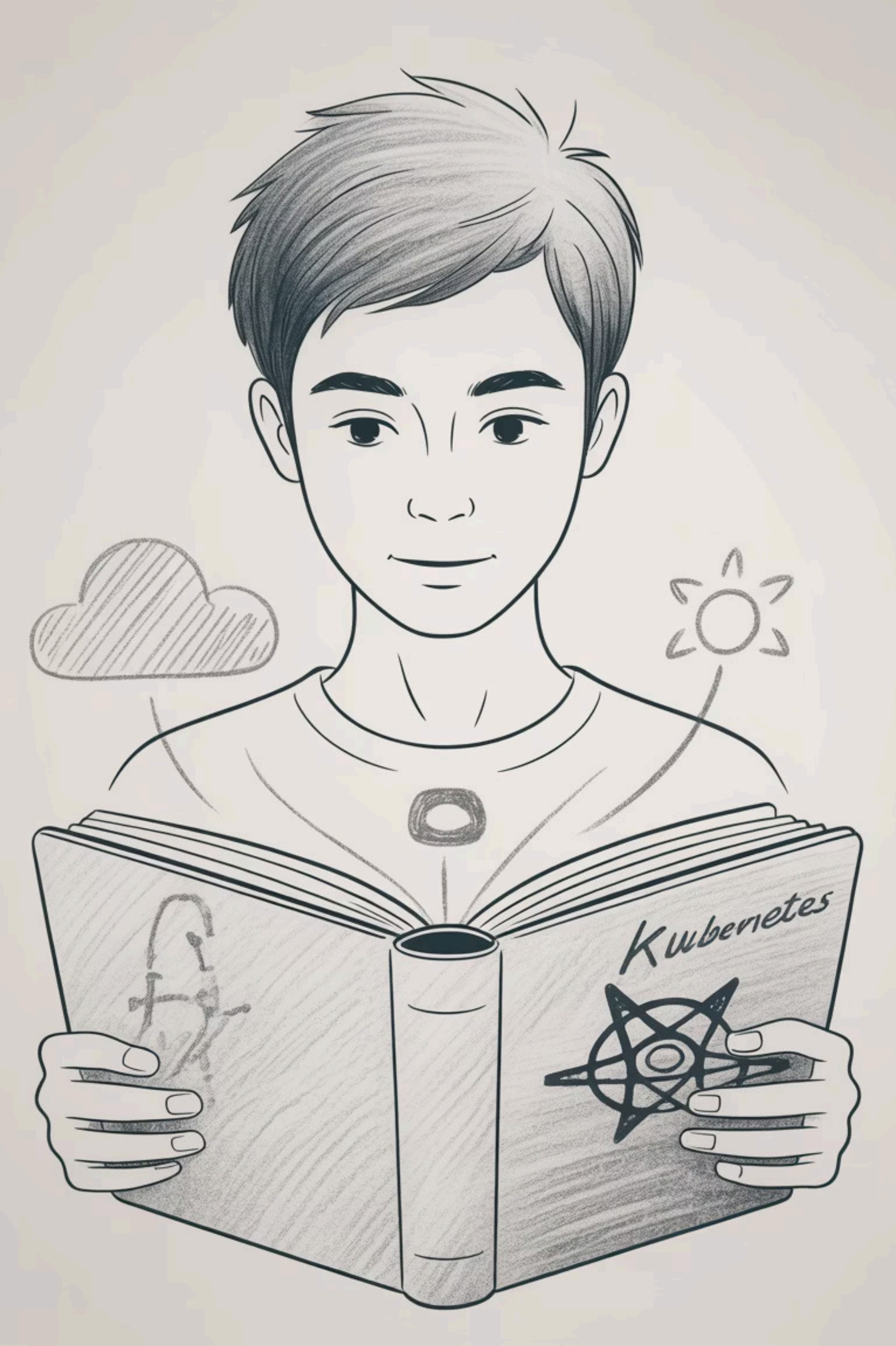
Scheduled on a node

04

Balanced traffic

05

Replaced if it fails

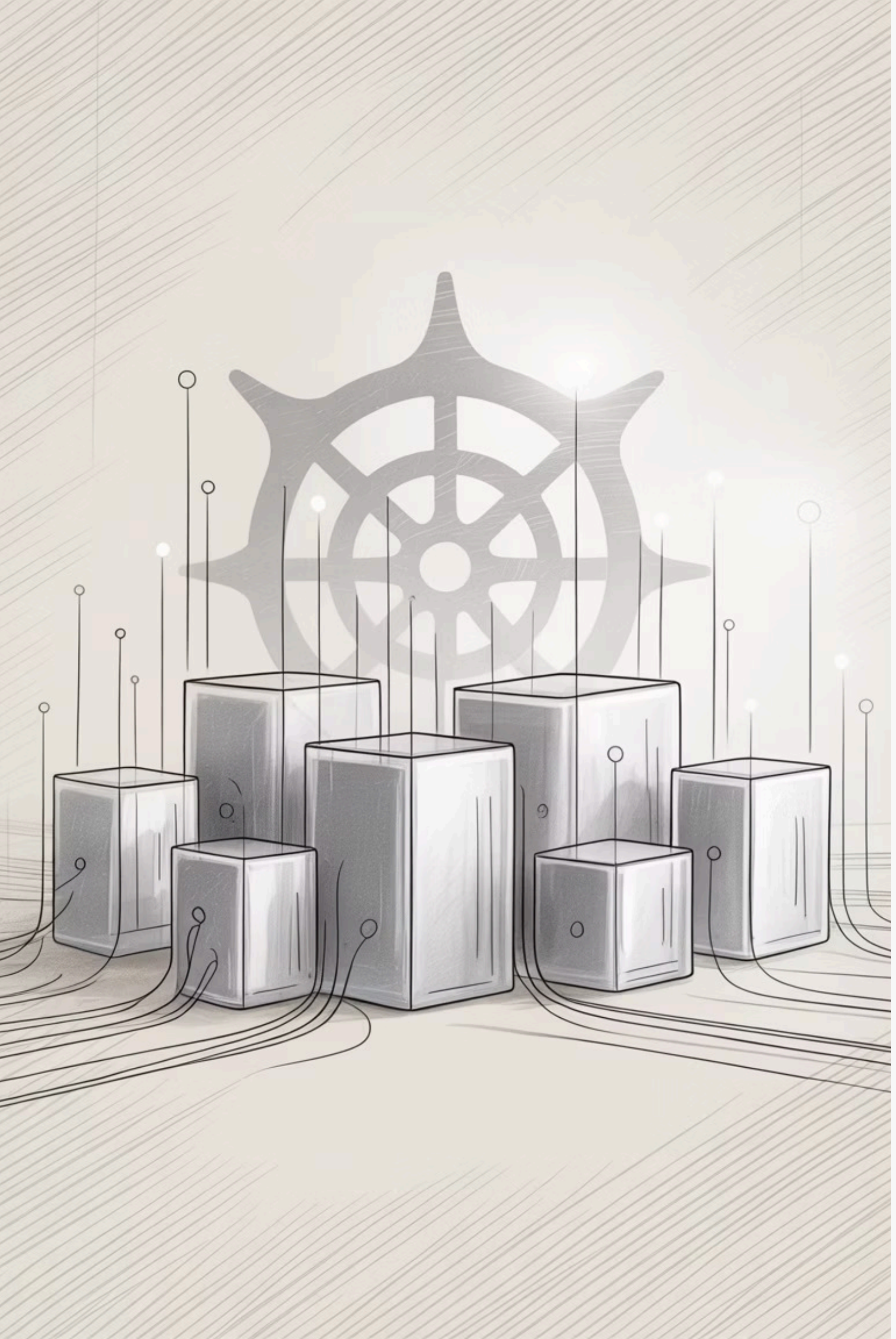


Benefits for You (as Freshers)

Job-ready skill

High demand in DevOps & Cloud

Learn how apps scale in production



Summary

Kubernetes = Container management platform

Automates deployment & recovery

Essential for cloud-native apps

Next Steps to Learn

Learn Docker basics

Try Minikube locally

Explore YAML configs

Join projects at work